

CO₂ - ASSESMENT TOOL 2

Scenario Based Questions

$$1) \quad I = \begin{bmatrix} 45 & 48 & 46 \\ 47 & 220 & 49 \\ 46 & 48 & 47 \end{bmatrix} \quad \text{mask} = \begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

$$T = 500$$

i,

$$R = (-45 - 48 - 46 - 47 - 49 - 46 - 48 - 47) + 220(8)$$

$$R = 1760 - 376$$

$$\boxed{R = 1384}$$

ii,

Given

$$T = 500$$

$$|R| = 1384 \geq 500 \quad (\because |R| \geq T)$$

$\therefore$ The center point is an isolated point

iii,

Defect Detection :- A high response indicates sharp, localized intensity discontinuity, flagging an anomaly like pixel defect.

Line detection

* It is a feature extraction technique to identify straight structures present in an image.

$$R = w * f(x, y)$$

Horizontal line mask :- The center row has +ve weights

$$H = \begin{bmatrix} -1 & -1 & -1 \\ 2 & 2 & 2 \\ -1 & -1 & -1 \end{bmatrix}$$

Vertical line mask :- The center column has +ve weight

$$V = \begin{bmatrix} -1 & 2 & -1 \\ -1 & 2 & -1 \\ -1 & 2 & -1 \end{bmatrix}$$

45 degree diagonal line :- Diagonal from top to right bottom has +ve weights

$$D_1 = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$$

-ve 45 degree diagonal line

$$D_2 = \begin{bmatrix} -1 & -1 & 2 \\ -1 & 2 & -1 \\ 2 & -1 & -1 \end{bmatrix}$$

Right top to left bottom has +ve weights.

$$R = \sum_{i=-1}^1 \sum_{j=-1}^1 w(i,j) f(x+i, y+j)$$

$$2) \quad I = \begin{bmatrix} 20 & 20 & 20 \\ 100 & 100 & 100 \\ 20 & 20 & 20 \end{bmatrix} \quad H = \begin{bmatrix} -1 & -1 & -1 \\ 2 & 2 & 2 \\ -1 & -1 & -1 \end{bmatrix}$$

- i, compute response of each mask
- ii, which mask produces highest response
- iii, what is orientation of detected line
- iv, Explain result in context of road land detection.

Horizontal

$$I = \begin{bmatrix} 20 & 20 & 20 \\ 100 & 100 & 100 \\ 20 & 20 & 20 \end{bmatrix} \quad H = \begin{bmatrix} -1 & -1 & -1 \\ 2 & 2 & 2 \\ -1 & -1 & -1 \end{bmatrix}$$

$$R = -20 - 20 - 20 + 200 + 200 + 200 - 20 - 20 - 20$$

$$R = 600 - 120$$

$$\boxed{R = 480}$$

Vertical

$$I = \begin{bmatrix} 20 & 20 & 20 \\ 100 & 100 & 100 \\ 20 & 20 & 20 \end{bmatrix} \quad V = \begin{bmatrix} -1 & 2 & -1 \\ -1 & 2 & -1 \\ -1 & 2 & -1 \end{bmatrix}$$

$$R = -20 + 40 - 20 - 100 + 200 - 100 - 20 + 40 - 20$$

$$\boxed{R = 0}$$

+45 degree diagonal

$$I = \begin{bmatrix} 20 & 20 & 20 \\ 100 & 100 & 100 \\ 20 & 20 & 20 \end{bmatrix} D_1 = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$$

$$R = 40 - 20 - 20 - 100 + 200 - 100 - 20 - 20 + 40$$

$$R = 0$$

-45 degree diagonal

$$I = \begin{bmatrix} 20 & 20 & 20 \\ 100 & 100 & 100 \\ 20 & 20 & 20 \end{bmatrix} D_2 = \begin{bmatrix} -1 & -1 & 2 \\ -1 & 2 & -1 \\ 2 & -1 & -1 \end{bmatrix}$$

$$R = -20 - 20 + 40 - 100 + 200 - 200 + 40 - 20 - 20$$

$$R = 0$$

ii. Horizontal line

iii. Horizontal line

iv. * In road lane detection, lane markings appears as bright pixels against a darker asphalt.

* The localization of a horizontal road lane marker within this section of camera frame

Gaussian Filtering

$$G(x, y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}}$$

Gradient Calculation

$$G = \sqrt{G_x^2 + G_y^2}$$

Non maximum Suppression

* remove unwanted pixels

Edge Tracking

* weak edges connected to strong

3/

Previous pixel center pixel Next pixel

42 60 35

• High Threshold (T_H) = 55

• Low Threshold (T_L) = 30

* Horizontal Gradient (G_x) = 36

* Vertical Gradient (G_y) = 48

iii) Gradient Magnitude

$$G = \sqrt{G_x^2 + G_y^2}$$
$$= \sqrt{1296 + 2304} = \sqrt{3600}$$

$$G = 60$$

iv) Non-maximum suppression

previous pixel = 42

next pixel = 35

* center pixel (60) > previous pixel (42)

* center pixel (60) > next pixel (35)

$\therefore$ central pixel survives non-maximum suppression.

iv) $G = 60$, $T_H = 55$, $T_L = 30$

condition: $G \geq T_H$ ($60 \geq 55$)

The magnitude is strictly greater than high threshold (T_H)

$\therefore$ Center pixel is automatically classified as strong edge pixel.

Road land boundary detection

Edge pixel

co-ordinates (x,y)

gradient direction

P₁

(10, 10)

44°

P₂

(11, 11)

46°

P₃

(12, 12)

45°

P₄

(18, 18)

92°

P₅

(19, 19)

91°

The vehicle links two edge pixel if Euclidean distance less than or equal to 2 pixel and diff in less than or equal to 5

Step 1:- Find distance b/w d₁ & d₂

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$d = \sqrt{(11 - 10)^2 + (11 - 10)^2} = \sqrt{2}$$

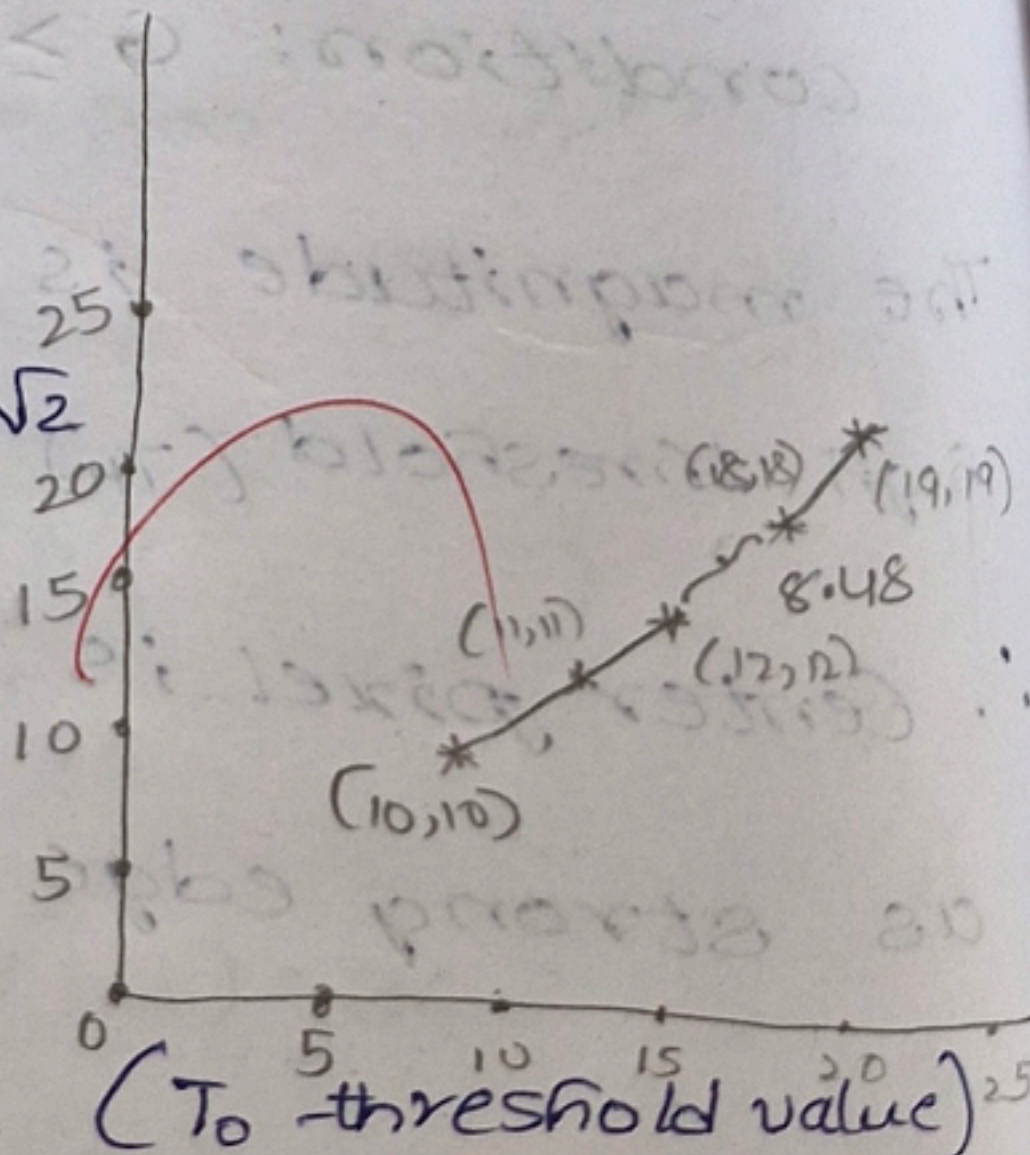
$$d = 1.414$$

Linking condition

$$|\theta_1 - \theta_2| < T_\theta$$

(T_θ threshold value)

$$|G_1 - G_2| < T_G$$



$$|46-44| = 2$$

$$1 \leq 2, \quad 2 \leq 5$$

P₂ to P₃

$$d = \sqrt{(12-11)^2 + (12-11)^2}$$

$$d = 1.414 \quad 1 \leq 2$$

$$|46-45| = 1$$

$$1 \leq 5$$

P₃ to P₄

$$d = \sqrt{36 + 36}$$

$$d = \sqrt{72}$$

$$d = 8.48$$

$$|45-92|$$

$$47$$

P₄ to P₅

$$d = \sqrt{(19-18)^2 + (19-18)^2} = \sqrt{2}$$

$$d = 1.414$$

$$|92-91| = 1 \leq 5$$

Scenario understanding (20%)	identification of issues (20%)	application (25%)	decision making (20%)	clarity of response (15%)	Total
20	20	25	20	15	100 W